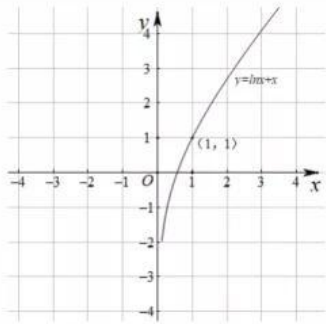
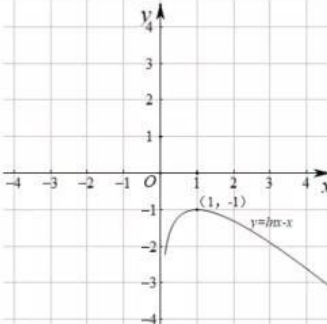
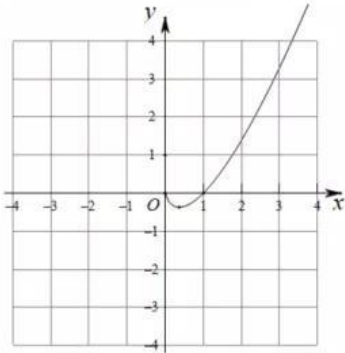
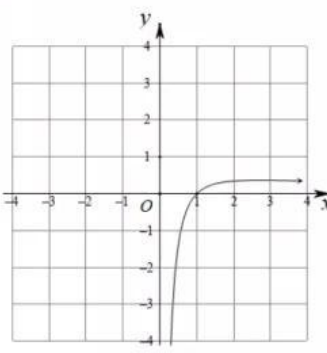
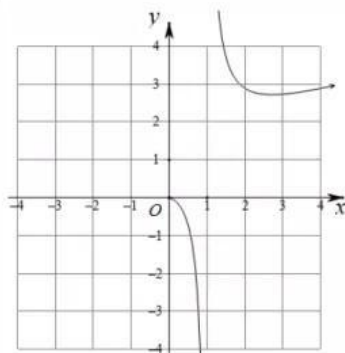
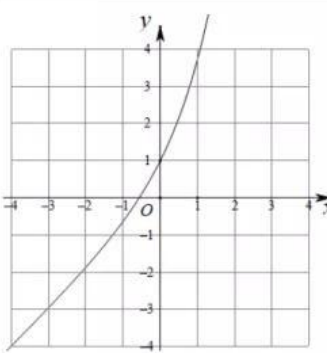
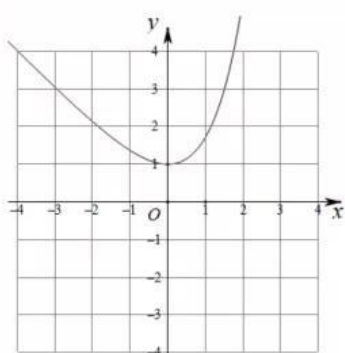
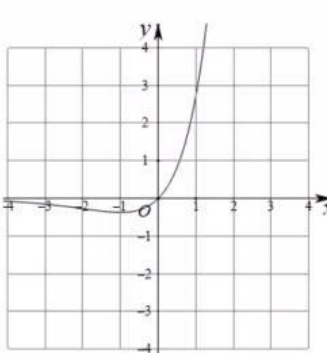
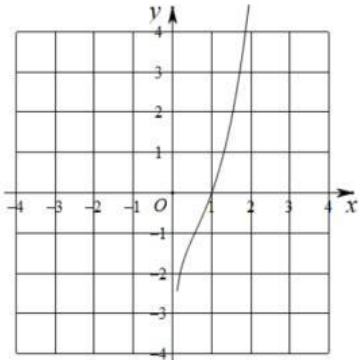
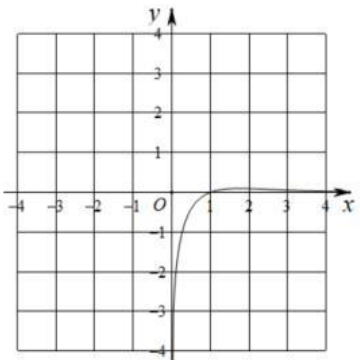
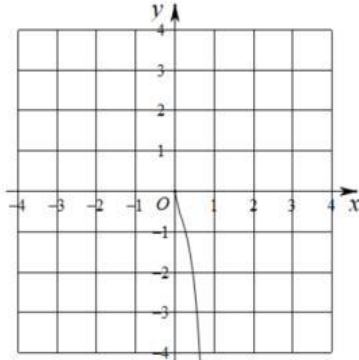
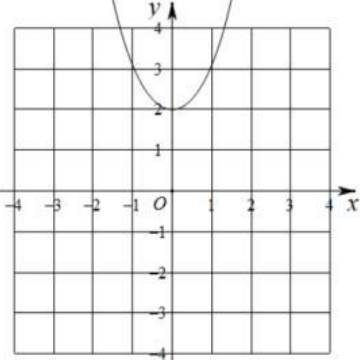
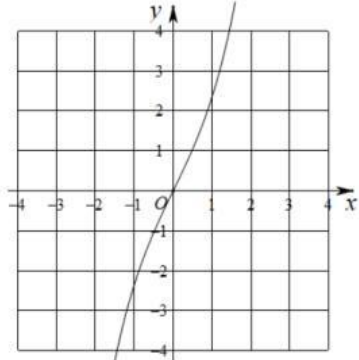
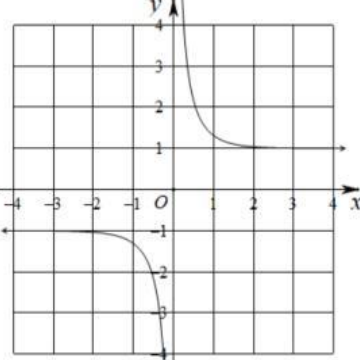
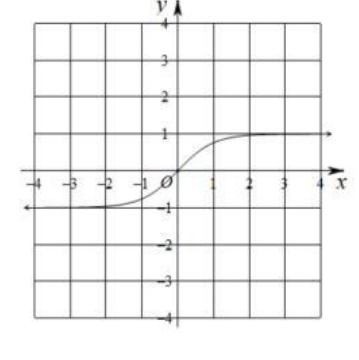
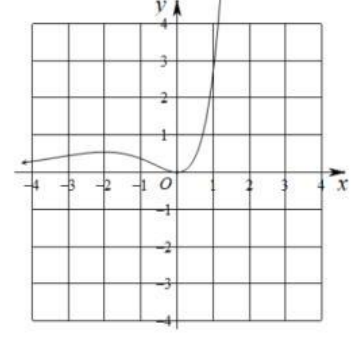
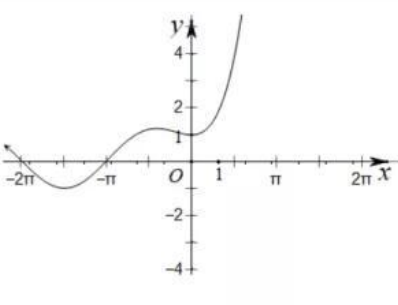
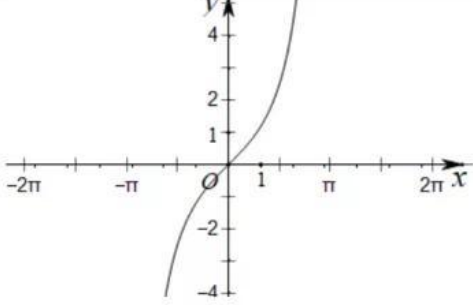
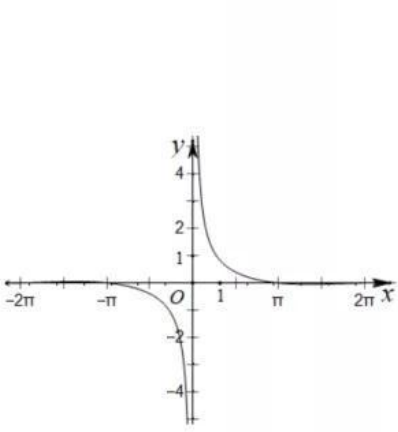
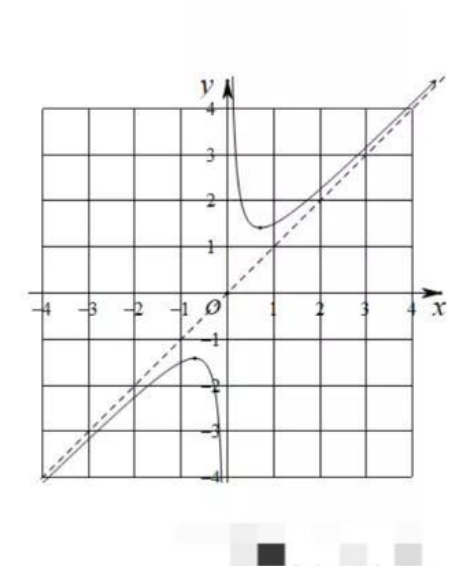
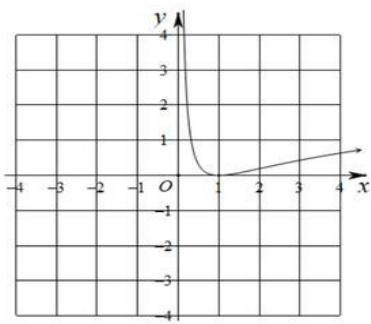
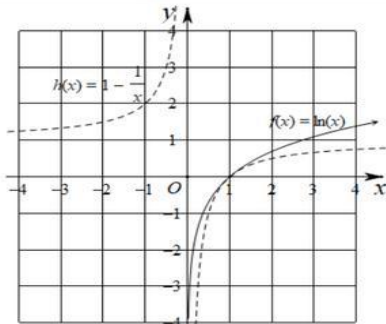
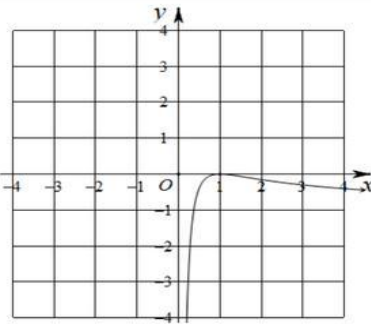
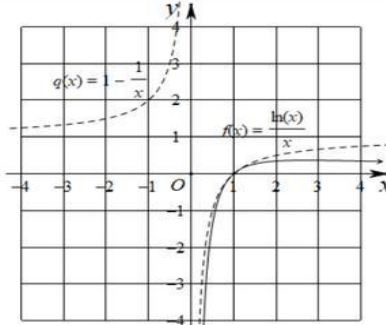
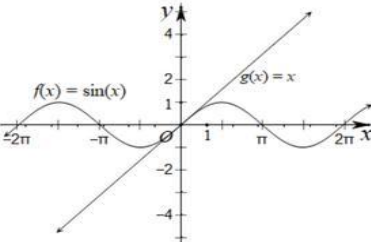
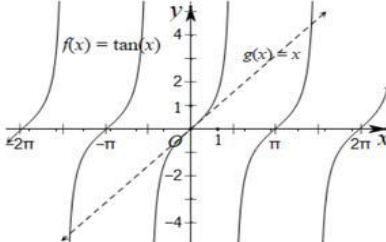


# 高中数学特殊函数图像

| 函数表达式                                                              | 图像                                                                                  | 函数表达式                                                           | 图像                                                                                    |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------|
| $y = \ln x + x$                                                    |    | $y = \ln x - x$                                                 |    |
| $y = x \ln x$<br>函数极值点<br>$\left(\frac{1}{e}, -\frac{1}{e}\right)$ |   | $y = \frac{\ln x}{x}$<br>函数极值点<br>$\left(e, \frac{1}{e}\right)$ |   |
| $y = \frac{x}{\ln x}$<br>函数极值点<br>$(e, e)$                         |  | $y = e^x + x$<br>过定点<br>$(0, 1)$                                |  |
| $y = e^x - x$<br>函数极值点<br>$(0, 1)$                                 |  | $y = xe^x$<br>函数极值点<br>$\left(-1, -\frac{1}{e}\right)$          |  |

|                                         |                                                                                     |                                         |                                                                                       |
|-----------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------|
| $y = e^x \ln x$                         |    | $y = \frac{\ln x}{e^x}$                 |    |
| $y = \frac{e^x}{\ln x}$                 |    | $y = e^x + e^{-x}$                      |    |
| $y = e^x - e^{-x}$                      |  | $y = \frac{e^x + e^{-x}}{e^x - e^{-x}}$ |  |
| $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ |  | $y = x^2 e^x$                           |  |
| 函数表达式                                   | 图像                                                                                  | 函数表达式                                   |  |

|                          |                                                                                   |                                                                                                                                                      |                                                                                      |
|--------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| $y = e^x - \sin x$       |  | $y = \frac{x^2}{\sin x}$                                                                                                                             |   |
| $y = \frac{\sin x}{x^2}$ |  | $y = x + \frac{k}{x}$<br>$k = \frac{1}{2}$<br>函数极值点<br>$\left(\frac{\sqrt{2}}{2}, \sqrt{2}\right),$<br>$\left(-\frac{\sqrt{2}}{2}, -\sqrt{2}\right)$ |  |

| 函数表达式                                                                                                                                                   | 图像                                                                                  | 函数表达式                                                                                                                                                       | 图像                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| $y = \ln x + \frac{1}{x} - 1$<br>函数过定点<br>$\left(\frac{\sqrt{2}}{2}, 0\right)$<br>$\left(-\frac{\sqrt{2}}{2}, 0\right)$<br>$\ln x \geq 1 - \frac{1}{x}$ |    | $y = \ln x + \frac{1}{x} - 1$<br>$\begin{cases} y = \ln x \\ y = 1 - \frac{1}{x} \end{cases}$<br>$\ln x \geq 1 - \frac{1}{x}$                               |    |
| $y = \frac{\ln x}{x} + \frac{1}{x} - 1$<br>函数极值点<br>$\left(\frac{1}{e}, -\frac{1}{e}\right)$<br>$\frac{\ln x}{x} \leq 1 - \frac{1}{x}$                  |   | $y = \frac{\ln x}{x} + \frac{1}{x} - 1$<br>$\begin{cases} y = \frac{\ln x}{x} \\ y = 1 - \frac{1}{x} \end{cases}$<br>$\frac{\ln x}{x} \leq 1 - \frac{1}{x}$ |   |
| $\begin{cases} y = \sin x \\ y = x \end{cases}$<br>函数极值点<br>$(e, e)$<br>$x > \sin x \quad x > 0$                                                        |  | $y = x - \tan x$<br>过定点<br>$(0, 1)$<br>$x < \tan x \quad 0 < x < \frac{\pi}{2}$                                                                             |  |

|                                                             |  |                                                                       |  |
|-------------------------------------------------------------|--|-----------------------------------------------------------------------|--|
| $y = \frac{e^x}{x}$ <p>函数极值点</p> <p><math>(1, e)</math></p> |  | $y = \frac{x}{e^x}$ <p>函数极值点</p> <p><math>(1, \frac{1}{e})</math></p> |  |
|-------------------------------------------------------------|--|-----------------------------------------------------------------------|--|

| 函数表达式                  | 图像 | 函数表达式                  | 图像 |
|------------------------|----|------------------------|----|
| $y = x + \sin x$       |    | $y = x - \sin x$       |    |
| $y = x \sin x$         |    | $y = \frac{x}{\sin x}$ |    |
| $y = \frac{\sin x}{x}$ |    | $y = x + \cos x$       |    |